

Population Genetics: Migration and Drift

Topic 1: Migration Topic 2: Genetic Drift (Founder Effect & Wright-Fisher Model)

These notes cover two major evolutionary forces: **Gene Flow (Migration)** and **Random Genetic Drift**. Understanding these is essential for evolutionary biology, conservation, and wildlife management – especially in the Indian context.

Topic 1: One Locus, Two Allele Model – Continent-Island Model of Migration

The Scenario

A large **continent** (source) with a stable allele frequency is exchanging migrants with a small **island** (recipient) that initially has a different frequency. Migration is one-way (continent $\rightarrow$ island). This models gene flow rescuing or swamping local adaptation.

Key Assumptions

- One locus, two alleles (e.g., A and a).
- Continent allele frequency for A = p_c (constant).
- Island allele frequency for A in generation t = p_t .
- Migration rate: m = proportion of island individuals that are new migrants from the continent each generation.
- No selection, mutation, or drift (island large enough to ignore drift).

Recurrence Equation

After one generation of migration:

$$p_{t+1} = (1 - m) \cdot p_t + m \cdot p_c$$

Interpretation:

- $(1 - m)$ fraction of islanders are non-migrants (carry p_t).
- m fraction are migrants (carry p_c).

Change in Allele Frequency Over One Generation (Due to Migration Alone)

$$\Delta p = m(p_c - p_t)$$

Where $\Delta p = p_{t+1} - p_t$.

Interpretation: The island moves toward the continent frequency by a fraction m of the current difference.

Equilibrium

The island frequency reaches equilibrium when $p_{t+1} = p_t$. Solving gives:

$$p_{eq} = p_c$$

Conclusion: Migration pulls the island allele frequency toward the continent's frequency. No permanent difference remains unless migration stops.

Approach to Equilibrium

The difference from the continent decays by factor $(1 - m)$ each generation:

$$p_t - p_c = (1 - m)^t(p_0 - p_c)$$

Topic 2: Effects of Small (Finite) Population Size

Sub-topic 1: Founder Effect

Definition: The loss of genetic variation and change in allele frequencies when a new population is started by a small number of individuals (founders) from a larger population.

Classic Example: Pingelap Atoll and Colorblindness

- **Event:** Typhoon in ~1775 reduced Pingelap island population to ~20 survivors.
- **The Founder:** One survivor carried a rare recessive allele for complete achromatopsia (total color blindness).
- **Result:** By chance, this allele became common. Today, 5-10% of Pingelap islanders have the condition (vs. <0.01% elsewhere).

Probability of Losing an Allele Due to Founder Effect If N founders are drawn from a large population where a neutral allele has frequency q :

$$P(\text{allele lost}) = (1 - q)^{2N}$$

Example: $q = 0.01$, $N = 5$ founders $\rightarrow (0.99)^{10} \approx 0.904$ (90% chance the allele is absent).

Note on factors affecting the probability of loss of an allele:

- **Number of founders (N):** Larger N reduces the probability of loss because more gene copies are sampled.
- **Initial frequency (q):** Higher q reduces the probability of loss because the allele is more likely to be included in the founder sample.

Empirical Study: White-eye Birds in Australia & New Zealand

- **Species:** Silveryeye (*Zosterops lateralis*).
- **Finding:** Each successive founding event (Australia $\rightarrow$ New Zealand $\rightarrow$ smaller islands) reduced genetic diversity (heterozygosity, number of alleles).
- **Conclusion:** Repeated founder effects cause stepwise loss of variation.

Sub-topic 2: Wright-Fisher Model of Genetic Drift

Verbal Description

- **Population:** Constant size N diploid individuals.
- **Life cycle:** Adults produce a huge pool of gametes. The next generation's $2N$ gene copies are drawn randomly from this pool.
- **No selection, mutation, or migration.** Only random sampling.
- **Consequence:** Allele frequencies fluctuate randomly each generation – this is genetic drift.

Important Results

$$H_t = H_0 \left(1 - \frac{1}{2N}\right)^t$$

- **Long-term outcome:** One allele becomes fixed (frequency 1), the other lost (frequency 0).
- **Mean frequency across populations:** $E[p_t] = p_0$ (unchanged).
- **Variance across populations:** Increases over time; at $t \rightarrow \infty$, $Var = p_0(1 - p_0)$.
- **Probability of fixation (neutral):** $= p_0$ (initial frequency).
- **Heterozygosity decay:** $H_t = H_0 \left(1 - \frac{1}{2N}\right)^t$

Buri's Experiment (1956) – Evidence for Drift

- **Organism:** Fruit fly (*Drosophila melanogaster*).
- **Setup:** 107 replicates, each $N = 16$, starting with 50% bw and 50% bw alleles.
- **Results over 19 generations:**
 - Allele frequencies diverged widely across replicates.
 - By generation 19: 30 replicates fixed bw, 28 fixed bw, rest still mixed.
 - Fixation probability ≈ 0.5 (matches initial frequency).
 - Heterozygosity declined every generation.
- **Conclusion:** Perfect experimental confirmation of Wright-Fisher drift.

Why Drift Matters: Conservation & Real-Life Examples

Genetic drift is not just a textbook concept – it has serious consequences for endangered species, especially when large populations become small and isolated due to human activity.

Example 1: Cheetahs (Historical Bottleneck – Drift in the Past)

- **History:** Cheetahs went through a severe population bottleneck ~10,000 years ago.
- **Result:** Modern cheetahs have very low genetic variation – so low that skin grafts between unrelated cheetahs are not rejected (they are nearly genetically identical).
- **Problem:** Low diversity makes them vulnerable to diseases and reduces fertility.

Example 2: Bengal Tigers in Central India – Habitat Fragmentation & Drift in Action

The Problem: From Connected to Isolated Central India once had continuous forests allowing tigers to move freely, mate across large areas, and maintain high genetic diversity. Today, this landscape (Madhya Pradesh, Maharashtra, Chhattisgarh) – which supports ~37% of India’s tiger population – is being rapidly fragmented by mining, roads, railways, and reservoirs.

What Drift Is Doing Now: The Similipal ”Black Tiger” Case In Similipal Tiger Reserve (Odisha), a real-time genetic crisis is unfolding:

- **The bottleneck:** In 2014, only 4 tigers remained (1 male, 3 females).
- **The recovery with a hidden problem:** The population grew to ~30 individuals, but all descended from the original 4 – every tiger is a close relative.
- **The drift signature:** A mutation causing pseudo-melanism (abnormally wide black stripes) has spread to ~60% of the population by chance.
- **Why this is dangerous:** This is a red flag that other, truly harmful recessive mutations are also spreading. When combined with inbreeding, this can have serious consequences.

The Conservation Response: Genetic Rescue Scientists used genetic data to identify a solution: introduce new tigers from a different population to restore gene flow. Two tigresses were translocated from Tadoba Tiger Reserve to Similipal in 2024-2025 – exactly the $\Delta p = m(p_c - p_t)$ equation in action.

Why Wildlife Corridors Matter Instead of emergency translocations, the long-term solution is maintaining natural corridors that allow tigers to move between reserves. Research confirms that tiger populations connected by forest corridors show higher gene flow and genetic diversity than isolated ones.

Key Takeaway: In small populations, drift overwhelms selection. India holds ~70% of the world's wild tigers. Their survival depends not just on poaching control, but on maintaining gene flow through corridors.

Gene flow (m) is the antidote to drift.

Comparison Table

Feature	Historical (Cheetah)	Bottleneck	Ongoing (Bengal Tiger – Similipal)	Fragmentation
Population size	Small, then recovered		Remains small & isolated	
Drift duration	Intense but temporary		Continuous	
Evidence of drift	Low diversity overall		"Black tiger" mutation in 60%	
Solution	Cannot reverse easily		Corridors + genetic rescue	

Remember: small populations drift, large populations resist drift, and migration connects them. The tiger in your local reserve needs corridors, not just counting.

Glossary / Definitions

Term	Definition
Allele frequency	Proportion of a specific allele in a population.
Bottleneck	Sharp reduction in population size, causing random loss of alleles.
Continent-island model	One-way migration from a large source to a small recipient.
Fixation	When an allele reaches frequency 1.0 in a population.
Founder effect	Genetic drift when a new population is started by a few individuals.
Gene flow	Movement of alleles between populations via migration.
Genetic drift	Random change in allele frequencies due to finite population size.
Heterozygosity (H)	Proportion of individuals that are heterozygous at a locus.
Migration rate (m)	Proportion of individuals that are new migrants each generation.
Neutral Allele	An allele that confers no selective advantage or disadvantage; its frequency changes only by drift, not by natural selection.
Wright-Fisher model	Mathematical model of drift in an idealised population.
Δp	Change in allele frequency over one generation.

Practice Problems (20 Questions)

Migration (Continent-Island)

1. In the continent-island model, if $p_c = 0.8$, $p_t = 0.3$, and $m = 0.1$, calculate Δp .
2. Using the data from Q1, calculate p_{t+1} .
3. If the island initially has $p_0 = 0$, $p_c = 1$, and $m = 0.05$, what is p_1 ? p_2 ?
4. An island has $p_t = 0.2$ and $p_c = 0.7$. What migration rate m would result in $\Delta p = 0.1$?
5. A continent has $p_c = 0.9$ for a beneficial allele. An island has $p_0 = 0.1$. After 10 generations of migration with $m = 0.1$, what is the expected island frequency?
6. True or False: In the continent-island model without drift, the island always reaches p_c exactly.
7. Explain in one sentence: Why does migration oppose local adaptation?

Founder Effect

8. A rare neutral allele has frequency $q = 0.005$ in a large population. If 10 individuals found a new population, what is the probability the allele is lost?
9. Why did colorblindness become common on Pingelap but not on neighboring islands?
10. A founder event starts with $N = 4$ individuals. What is the probability that a neutral allele with frequency $q = 0.1$ in the source population is present in the founders?
11. Explain how successive founding events (like in white-eye birds) reduce genetic diversity.

Wright-Fisher Drift

12. In the Wright-Fisher model, what is the probability that a neutral allele with initial frequency 0.3 eventually becomes fixed?
13. A population of $N = 50$ starts with heterozygosity $H_0 = 0.5$. What is expected H after 100 generations?
14. Buri's experiment used $N = 16$ flies per replicate. What fraction of heterozygosity remained after 1 generation? After 10 generations?
15. True or False: In the Wright-Fisher model, the mean allele frequency across many replicate populations changes over time.
16. Why does variance in allele frequency increase over time under drift?

17. A population of size $N = 10$ starts with two neutral alleles at equal frequency (0.5 each). What is the probability that after many generations, the population is fixed for one allele? For the other?
18. Explain in one sentence: How does population size affect the rate of heterozygosity loss?

Conservation & Integration

19. In the Similipal tiger example, explain how the "black tiger" mutation spreading to 60% of the population is a signature of genetic drift, not natural selection.
20. Compare and contrast: How do migration (gene flow) and drift have opposite effects on genetic variation between populations?

Answer Key

1. $\Delta p = m(p_c - p_t) = 0.1(0.8 - 0.3) = 0.1 \times 0.5 = 0.05$
2. $p_{t+1} = p_t + \Delta p = 0.3 + 0.05 = 0.35$
3. $p_1 = (1-0.05) \times 0 + 0.05 \times 1 = 0.05$; $p_2 = (1-0.05) \times 0.05 + 0.05 \times 1 = 0.0475 + 0.05 = 0.0975$
4. $\Delta p = m(p_c - p_t) \rightarrow 0.1 = m(0.7 - 0.2) \rightarrow 0.1 = m(0.5) \rightarrow m = 0.1/0.5 = 0.2$
5. $p_t = p_c + (p_0 - p_c)(1 - m)^t = 0.9 + (0.1 - 0.9)(0.9)^{10} = 0.9 + (-0.8)(0.3487) = 0.9 - 0.279 = 0.621$
6. True (in the deterministic model with no drift, equilibrium is exactly p_c).
7. Migration brings in alleles from the continent that may be maladapted to local island conditions, swamping local beneficial alleles.
8. $P(\text{loss}) = (1 - q)^{2N} = (0.995)^{20} \approx 0.9046$ (90.5% chance of loss).
9. By chance, the typhoon survivors (founders) included a carrier of the allele. Subsequent drift and inbreeding increased its frequency – a founder effect.
10. $P(\text{present}) = 1 - (1 - 0.1)^8 = 1 - (0.9)^8 = 1 - 0.4305 = 0.5695$ (~57%).
11. Each founding event samples a small subset of the previous population's alleles. Rare alleles are easily lost by chance, reducing total diversity stepwise.
12. Probability of fixation = initial frequency = 0.3 (or 30%).
13. $H_{100} = 0.5 \times (1 - 1/100)^{100} = 0.5 \times (0.99)^{100} \approx 0.5 \times 0.366 = 0.183$
14. Per generation fraction remaining = $1 - 1/(2N) = 1 - 1/32 = 0.96875$. After 10 generations: $0.96875^{10} \approx 0.727$ (72.7% of heterozygosity remains).
15. False. The mean frequency remains constant at p_0 .
16. Replicate populations wander randomly; over time some fix one allele, some fix the other, so variance across replicates increases.
17. Probability fixed for one allele = 0.5; fixed for the other = 0.5. (Total fixation probability = 1, split equally.)
18. Smaller populations lose heterozygosity faster because $1/(2N)$ is larger.
19. The black coat mutation is neutral (no known survival advantage or disadvantage). Its rise to 60% is due to random sampling in a tiny, inbred population – exactly the Wright-Fisher prediction that drift can change frequencies of neutral alleles. If selection were driving it, we would expect a fitness difference, which has not been demonstrated.
20. Migration reduces genetic differences between populations (homogenizes). Drift increases differences between populations by randomly fixing different alleles in each.